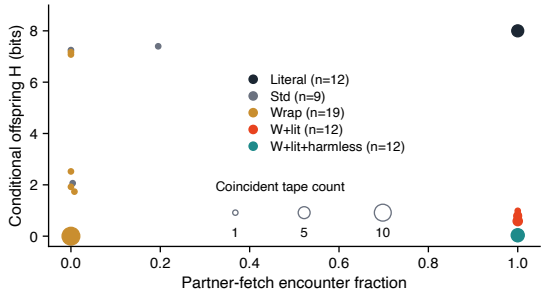
a BFF contracts: detection and persistence

Variant	Worlds	Modal detected	Pop. detected	Later collapse
Standard	24	9	7	0
Literal, no wrap	12	12	5	5
Standard, harmless	12	7	3	0
Pointer wrap	24	19	19	0
Wrap + literal	12	12	12	12
Wrap + literal, harmless	12	12	12	0

Literal emits operands; harmless = unmatched brackets do not halt
Modal generation-2 normalized culture gain ≥0.3; pop ≥50% heritable
Later collapse = terminal share <10% after population detection

b First copiers: fetch versus partner inflow



64 first-modal tapes, one/world; circle area = coincident count
H = offspring entropy conditional on parent, random partners
Standard harmless subset has no archived H panel

c Bandwidth controls: open fetch, zero inflow

Operand copies r	Worlds	H bits	Fetch fraction
1	12	8.0	1.00
2	6	0.0	1.00
3	6	0.0	1.00

Literal P a b emits a,b repeated r times

World medians of conditional H and partner-fetch fraction
r≥2: zero partner inflow with fetch in every encounter

d Halting dial: all worlds included in counts

Halt p	Worlds	P tapes persist	All-world median final P share
0	12	12	0.984
0.01	6	6	0.984
0.03	6	6	0.984
0.1	6	6	0.984
0.3	6	5	0.102
1	12	0	0.002

p: unmatched bracket halts; P = literal-copy instruction
Persistence: P-tape share ≥10% at epoch 16,384; final share median

e First / final inflow and fetch

Variant	n worlds	H bits first / final	Fetch fraction first / final
Literal, no wrap	12	7.99 / 7.99	1.00 / 1.00
Standard	9	0.04 / 0.00	0.00 / 0.00
Pointer wrap	19	0.00 / 0.00	0.00 / 0.00
Wrap + literal	12	0.75 / 0.64	1.00 / 1.00
Wrap + literal, harmless	12	0.04 / 0.00	1.00 / 1.00

First = first modal-life tape; final = terminal modal tape

Paired world sampling; values are medians across worlds
Fetch fraction and conditional H use the same units as b,c

f Registered individuality verdicts

Claim (bits)	Worlds pass/n	Verdict
Q1 Z80 first H>1	80/80	met
Q2 Z80 loop final H<0.5	63/67	met
Q3 Z80 H drop ≥2	63/80	met
Q4a BFF standard first H<0.5	6/9	between
Q4b BFF literal first H>1	12/36	rejected

Q1–3 Z80 G; Q2 only 67 loop-bearing finals; Q4a standard BFF
Q4b pools literal/no-wrap, wrap+literal, wrap+literal/harmless (36)
Support ≥90% (Q3 ≥75%); reject ≤60% (Q3 ≤50%); else between